

Secure Automated Web Architecture

Description

This repository provisions a secure, automated web architecture on AWS using Infrastructure as Code (IaC). It automates the deployment of a hardened VPC, isolated public subnet, custom security group rules, and an EC2 web server while enforcing security best practices.

Technologies Used

- **AWS:** Hosting VPC, Subnet, Internet Gateway, Security Groups, and EC2 instance.
- **Terraform:** Infrastructure as Code (IaC) tool used to declare and provision resources.
- **GitHub Actions:** Continuous Integration/Continuous Deployment (CI/CD) pipeline management.
- **tfsec:** Static analysis security scanner for Terraform configurations.

Architecture

The infrastructure is built using a defense-in-depth approach: * **VPC & Subnet Isolation:** Resources are isolated within a 10.0.0.0/16 Virtual Private Cloud with a public subnet (10.0.1.0/24) attached to an Internet Gateway for routed access. * **Security Group Lockdown:** Inbound access via `tkh-web-security-group` restricts administrative access. HTTP (port 80) traffic is open for public web access, while SSH (port 22) access is restricted to authorized administrative IPs.